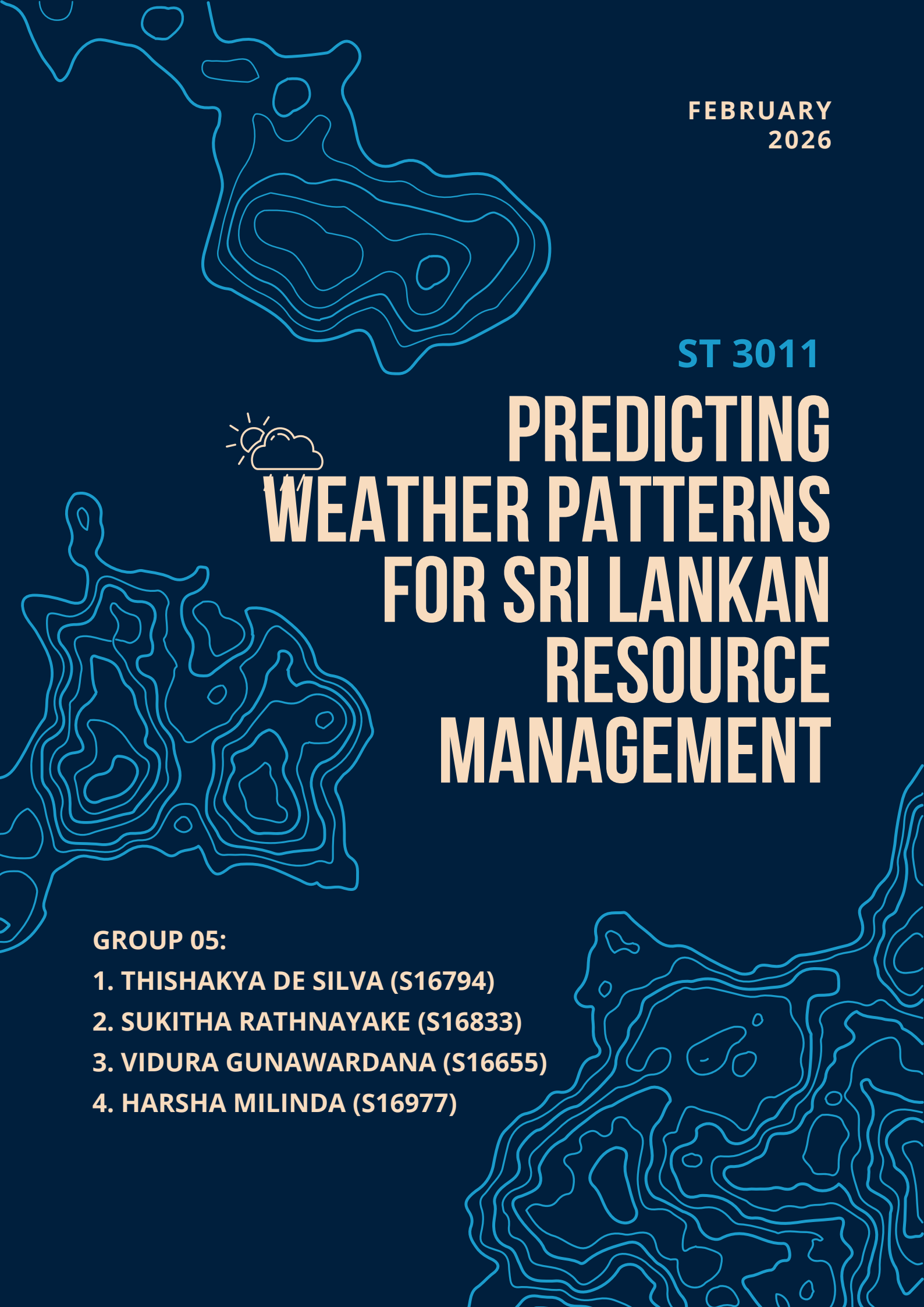




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ST 3011



PREDICTING WEATHER PATTERNS FOR SRI LANKAN RESOURCE MANAGEMENT

GROUP 05:

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Abstract

This study aims to support effective resource management in Sri Lanka by analyzing and modeling key weather patterns that influence water availability, agriculture, energy systems, and tourism. The first objective focuses on understanding and predicting daily precipitation through exploratory analysis of rainfall distributions, precipitation hours, and temporal patterns, followed by statistical hypothesis testing to compare monsoon and inter-monsoon rainfall, confidence interval estimation for the wettest cities, and logistic regression modeling to predict rainfall occurrence. The second objective examines apparent temperature and thermal comfort by comparing air temperature with apparent temperature across cities, analyzing annual cycles, testing elevation-based differences using ANOVA, assessing correlations with meteorological predictors, and developing a multiple linear regression model to predict apparent temperature. The third objective investigates wind dynamics and evapotranspiration by exploring wind speed, gusts, dominant wind directions, and seasonal evapotranspiration patterns, and by applying time-series forecasting to predict future evapotranspiration for a selected city. The final objective assesses the geographic influence on climate variability by identifying extreme temperature and wind events, visualizing spatial relationships between elevation and key climate variables, and constructing regression models to quantify how latitude, longitude, and elevation influence mean annual shortwave radiation, thereby providing insights into the spatial drivers of Sri Lanka's climate system.

1. Description of the Dataset

Dataset name: Sri Lanka Weather Dataset

Link: <https://www.kaggle.com/datasets/rasulmah/sri-lanka-weather-dataset>

The Sri Lanka Weather Dataset is a comprehensive collection of weather data for 30 prominent cities in Sri Lanka, covering the period from January 1, 2010, to January 1, 2023. The dataset offers a wide range of meteorological parameters, enabling detailed analysis and insights into the climate patterns of different regions in Sri Lanka.

Variable Name	Type	Description
Time	Time	The timestamp of each weather observation starting from 2010-01-01 to 2023-06-17
Weather Code	Qualitative (Nominal)	A numerical code representing the weather conditions at the given time
Temperature	Quantitative	Maximum, minimum, and mean values of 2-meter temperature
Apparent Temperature	Quantitative	Maximum, minimum, and mean values of apparent temperature, which takes into account factors like wind chill or heat index
Sunrise and Sunset	Quantitative	The times of sunrise and sunset for each day.
Shortwave Radiation	Quantitative	Sum of shortwave radiation received during the observation period.
Precipitation	Quantitative	Total sum of precipitation, including rainfall and snowfall.
Precipitation Hours	Quantitative	The duration of time with measurable precipitation.
Wind Speed and Gusts	Quantitative	Maximum values of wind speed and wind gusts at 10 meters above ground level.

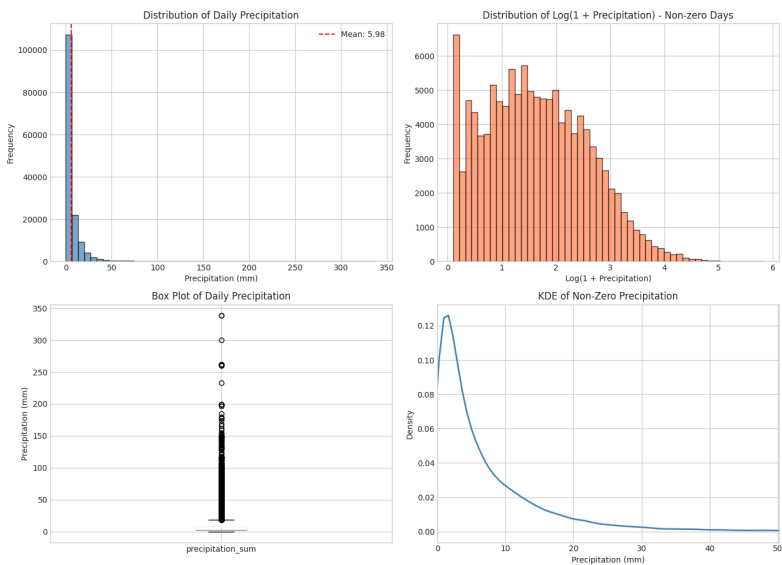
Wind Direction	Qualitative (Nominal)	Dominant wind direction at 10 meters above ground level.
Evapotranspiration	Quantitative	Reference evapotranspiration (ET0) based on the FAO Penman-Monteith equation.
Latitude, Longitude, Elevation	Quantitative	Geographic coordinates and elevation of each city.
Country and City	Qualitative (Nominal)	Names of the country and city corresponding to each weather observation.

2. Data Pre-processing

Before conducting the EDA, the dataset was carefully cleaned and checked for quality.

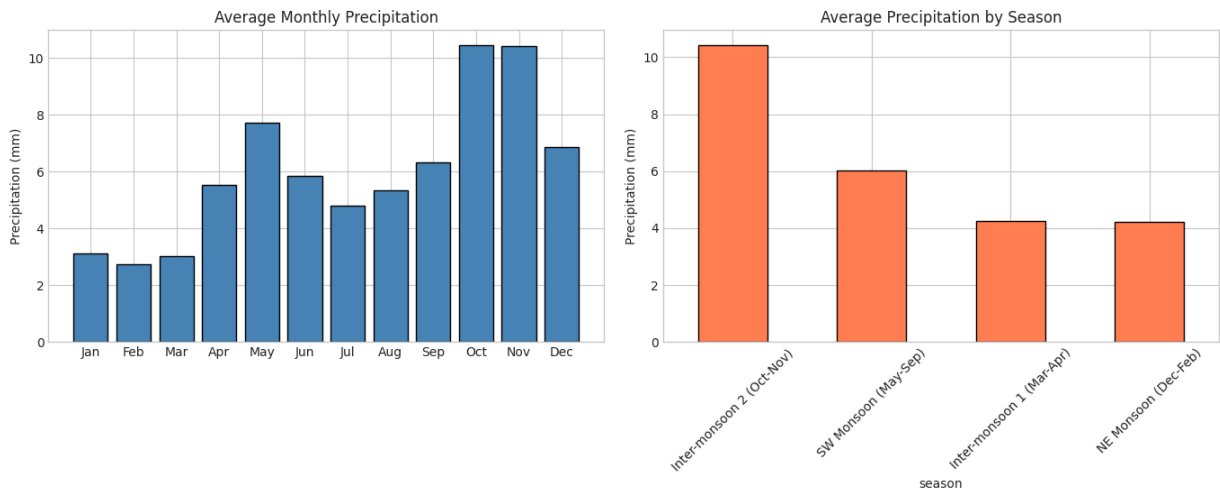
- No missing values and duplicates were found.
- All variables were checked for appropriate data types, and no changes were required.
- The data was screened for unrealistic or inconsistent values, but none were detected.

3. Main Results of Descriptive Analysis



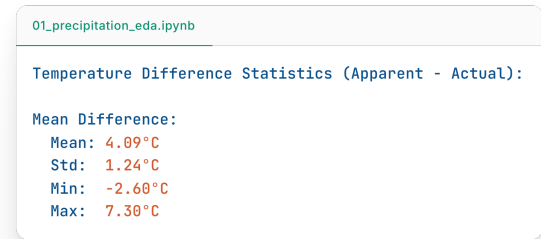
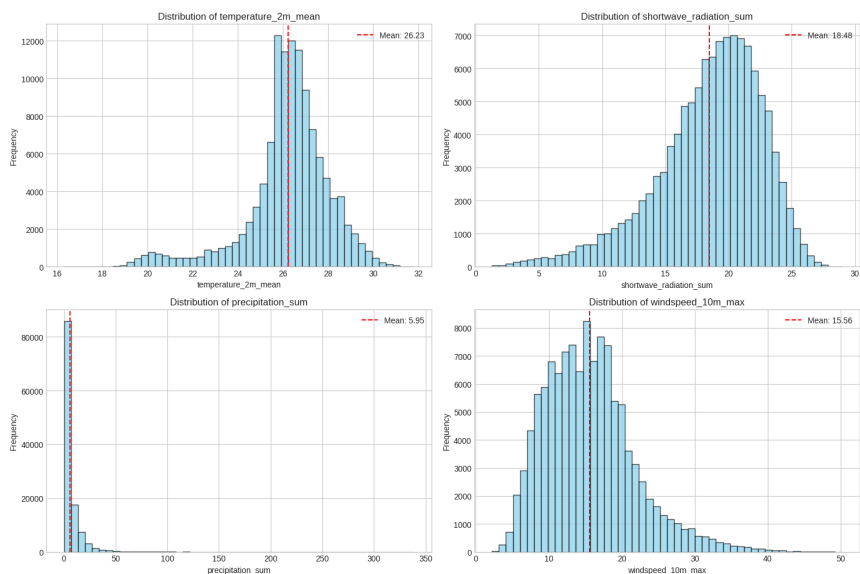
The precipitation distribution plots show that 18.1% of days have zero rainfall and that most rainfall events are relatively small. The histogram and density curve reveal a strongly right-skewed distribution, with a mean daily precipitation of 5.98 mm and a lower median of 2.60 mm. The long tail in the plot reflects extreme rainfall events, with a maximum of 338.80 mm, consistent with the high skewness value of 5.99.

The boxplot and IQR-based outlier plot highlight the presence of heavy rainfall extremes. With Q1 at 0.40 mm and Q3 at 7.50 mm, values above 18.15 mm appear as



numerous outliers beyond the whiskers. A total of 11,145 observations (7.56% of the data) are classified as outliers, visually emphasizing that intense rainfall events, though infrequent, are a key feature of the dataset.

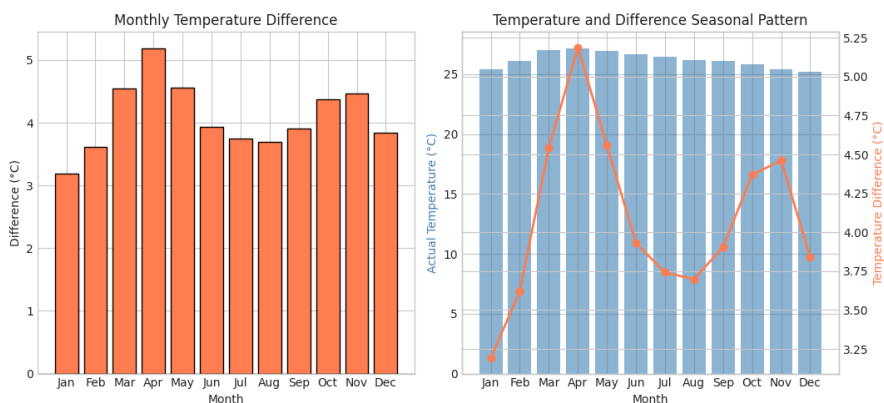
The monthly and seasonal time-series and bar plots of mean daily precipitation across climatic seasons reveal clear temporal variability. As depicted in the seasonal comparison plot, the second inter-monsoon period (October–November) exhibits the highest average daily precipitation at 10.43 mm, standing out prominently above other seasons. The southwest monsoon (May–September) follows with a mean of 6.02 mm per day, while both the first inter-monsoon period (March–April) and the northeast monsoon (December–February) show comparatively lower averages of 4.26 mm and 4.22 mm, respectively. These patterns, clearly visible in the plots, reflect the dominant influence of monsoonal circulation on rainfall distribution throughout the year.



The Temperature and Shortwave Radiation plots both exhibit a negative (left) skew, indicating that while values frequently cluster at the higher end (averaging 26.23°C and 18.48 units, respectively), there are infrequent occurrences of significantly lower values. In stark contrast, the Precipitation plot shows an extreme positive (right) skew, where the vast majority of data points sit near zero, though a long tail reveals rare, high-intensity events that pull the mean to 5.95. Similarly, the Windspeed distribution is positively skewed, with most observations concentrated between 10 and 20 units and a mean of 15.56.

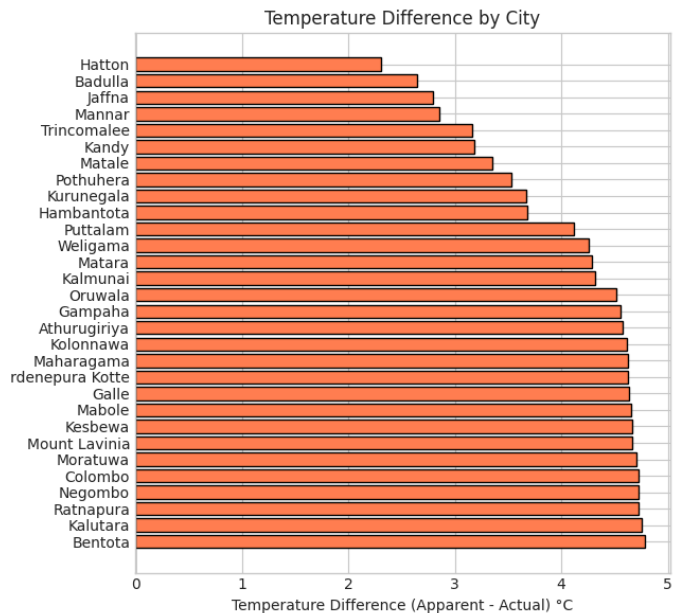
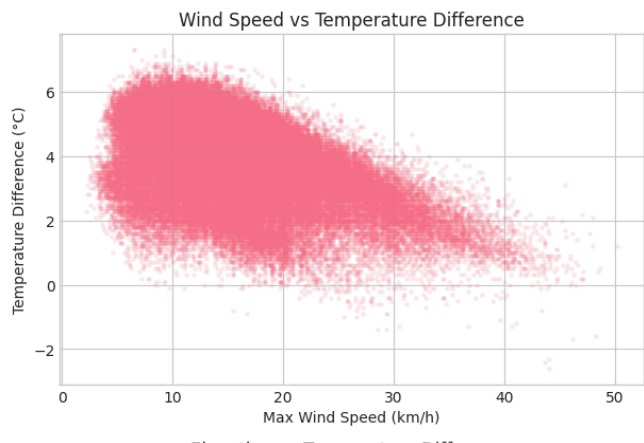
The mean difference of 4.09°C might sound small, but it is the difference between Comfortable and Extreme Caution. Because actual temperatures are already high (26-27°C), the extra 4°C of apparent heat is driven primarily by humidity.

In tropical climates like Sri Lanka, high moisture prevents sweat from evaporating. When your body cannot shed heat, your internal temperature rises, leading to the sweltering feeling. The massive spike in April aligns with the inter-monsoon period. During this time, the sun is directly overhead, wind speeds drop, and humidity skyrockets.



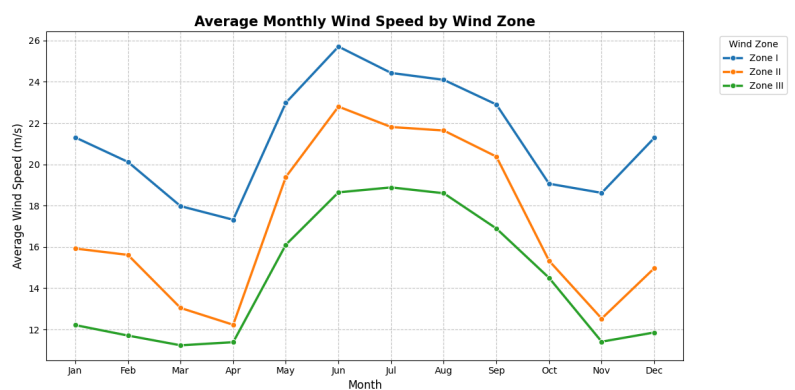
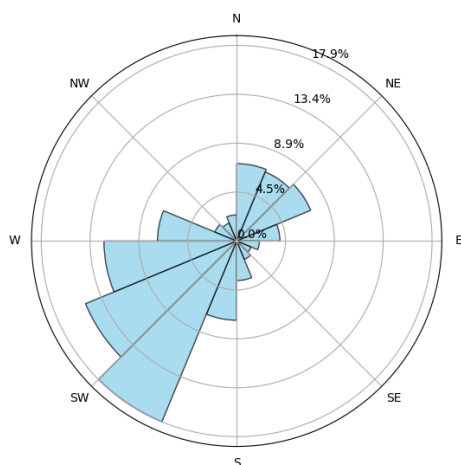
The barplot shows that Hatton has the lowest temperature gap (2.31°C) because cooler, thinner air at high altitudes cannot hold as much moisture. While cities like Bentota and Kalutara have the highest gaps (+4.78°C). Even though they are near the ocean, the combination of intense tropical sun and high sea-level humidity creates a steam room effect.

The only relief for these cities is the sea breeze, which is why we see a negative correlation with wind speed in the scatter plot (-0.50) without wind, these coastal areas would be nearly uninhabitable during peak hours.3. Urban vs. Rural Heat Stress. While the data doesn't explicitly label urban vs. rural, we can infer the impact of the Urban Heat Island (UHI) effect.Heat Persistence: In temcorr.png, notice how the apparent temperature stays



high even when actual temperatures are moderate. In urban centers like Colombo or Galle, concrete and asphalt absorb heat during the day and radiate it back at night.The Risk Factor: This means that in urban coastal areas, the apparent temperature doesn't just peak higher it stays higher for longer, depriving the human body of the nighttime cooling needed to recover from heat stress.

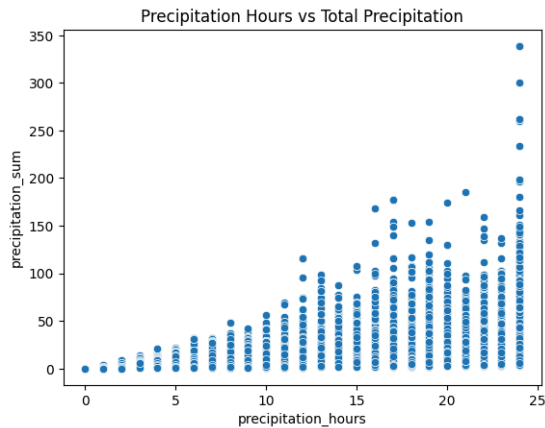
Overall Dominant Wind Direction Frequency (Sri Lanka)



Wind power on the island is primarily generated from two seasonal monsoon directions: the Southwest and the Northeast. The Southwest wind is notably more significant, accounting for over 18% of the recorded wind time, suggesting it is both stronger and more frequent. Wind speed data is classified into zones based on measured speeds. Zone I represents the highest speeds, with averages peaking around 25-26 m/s. In contrast, Zone III recorded the slowest average wind speed around 10-12 m/s.

4. Advanced Analysis

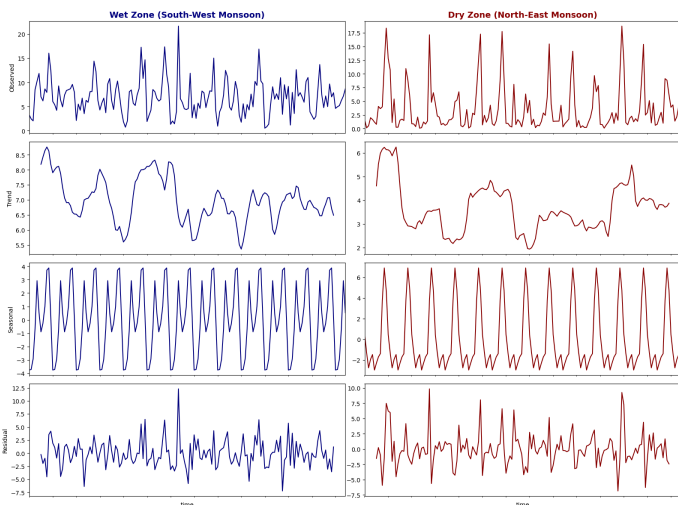
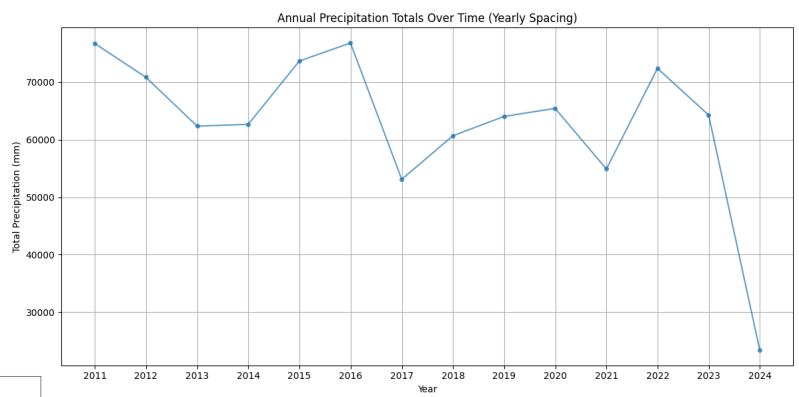
4.1. Objective 1: Understanding and Predicting Daily Precipitation



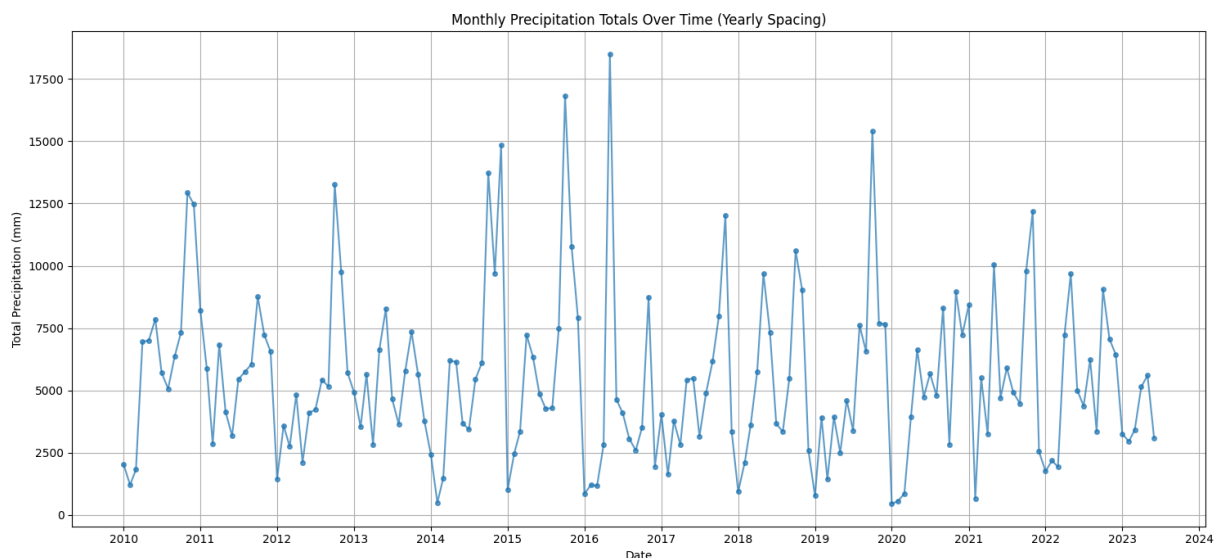
Through the advanced analysis we aim to create a logistic model that can accurately predict the occurrence of rainfall. First we dropped variables that are not important to our analysis ('weathercode', 'sunrise', 'sunset', 'country', 'rain_sum', 'snowfall_sum')

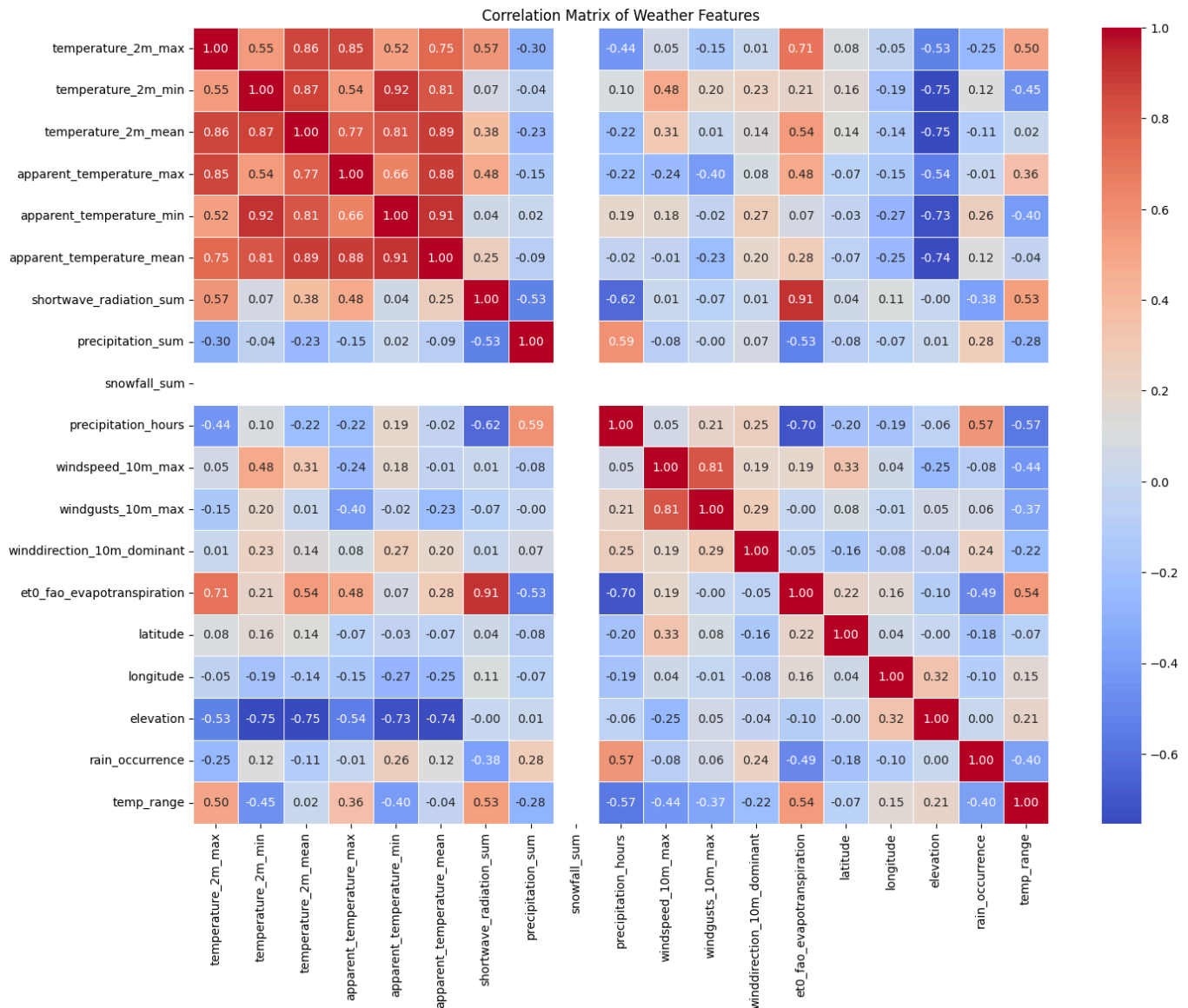
By observing the graph on the left it is clear that as the precipitation hours increase, the total precipitation also increases.

By looking at the graph on the right it is clear that 2016 is the year with the highest annual rainfall; this aligns with the fact that it was the year where Sri Lanka received the highest rainfall in over 18 years, and major floods.



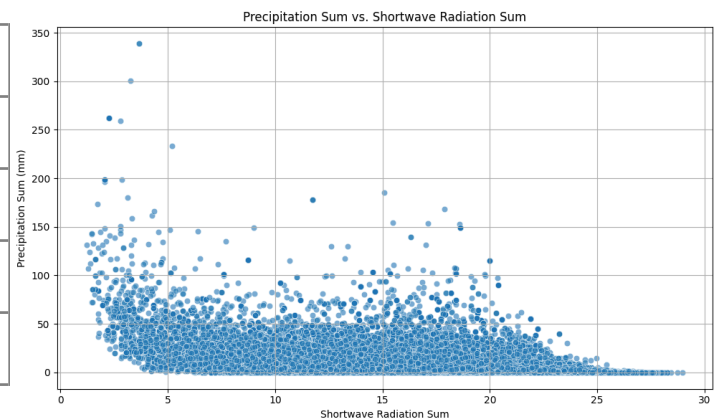
By observing the monthly precipitation over time, it is clear that there is an overall seasonality, and by further observing the trends separately for the Wet Zone and the dry zone at their monsoon seasons, we can further see the seasonality.





It is clear that there is a high correlation between the many temperature measurements, and between the wind speed as well as windgusts, therefore when attempting to fit a machine learning model we can drop some of these. Apart from that we can discard the snowfall column since it is irrelevant when doing our analysis. By looking at the correlation coefficients of the predictors with the target variable it is clear that the hours of precipitation has the highest positive correlation while the shortwave radiation has the highest negative correlation. This negative relationship is further shown in the scatter plot below.

	precipitation_sum
precipitation_hours	0.590132
temperature_2m_max	-0.298439
et0_fao_evapotranspiration	-0.525634
shortwave_radiation_sum	-0.533051



In order to investigate if there is a significant difference in the rainfall of the 4 different seasons(North East Monsoon, South West Monsoon, and the 2 inter- monsoon seasons), we conducted an ANOVA test, and found that there is a significant difference in at least one season, therefore we utilized Tukey’s test to explore which seasons have a significant difference.

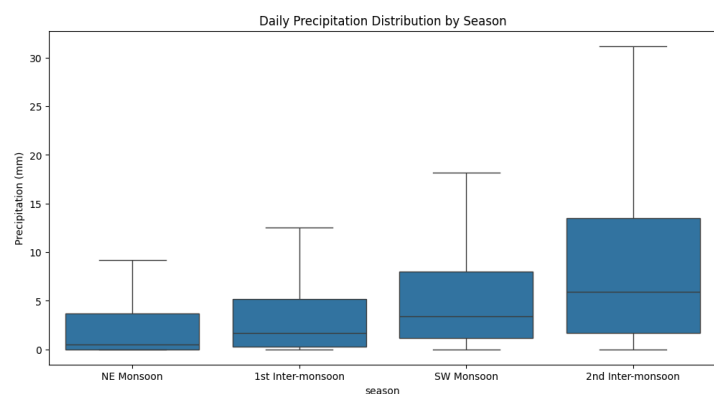
```

=== One-Way ANOVA Results ===
F-Statistic: 2209.1638
P-Value: 0.0000e+00

Significant difference found. Running Tukey's HSD...
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1      group2      meandiff p-adj    lower    upper    reject
-----
1st Inter-monsoon 2nd Inter-monsoon    6.1679    0.0    5.9367    6.3991    True
1st Inter-monsoon    NE Monsoon   -0.0463  0.9411   -0.255    0.1624   False
1st Inter-monsoon    SW Monsoon    1.7628    0.0    1.5717    1.9539    True
2nd Inter-monsoon    NE Monsoon   -6.2142    0.0   -6.4276   -6.0008    True
2nd Inter-monsoon    SW Monsoon   -4.4051    0.0   -4.6013   -4.2089    True
    NE Monsoon    SW Monsoon    1.8091    0.0    1.6399    1.9783    True
-----

```

By looking at the results it is clear that apart from the 1st inter monsoon and North East monsoon, all the other monsoon seasons have a total precipitation which is significantly different from each other



This is further exemplified in the boxplot of precipitation drawn by season, which also further shows the 2nd Inter monsoon season has the highest total rainfall.

dickey_fully.ipynb

```

=== Augmented Dickey-Fuller Test Results ===
H0: The time series is non-stationary
H1: The time series is stationary

ADF Statistic: -37.8355
P-Value: 0.0000e+00
Critical Values:
  1%: -3.4304
  5%: -2.8616
 10%: -2.5668
Result: STATIONARY (Reject H0)
The rainfall data does not have a trend.
The mean and variance are constant over time.

```

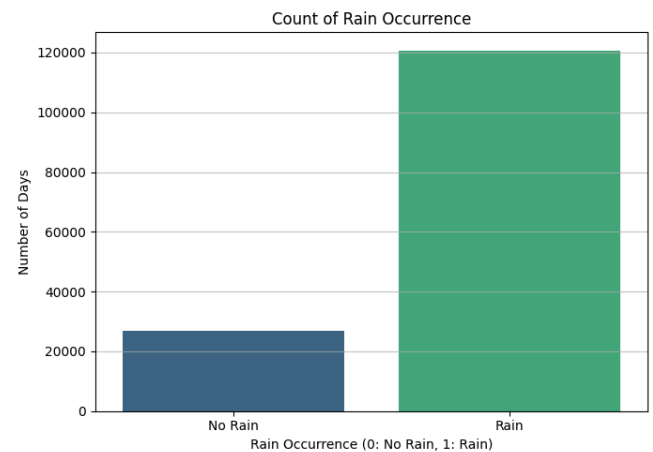
To test the stationarity, we also utilized the Augmented Dickey Fuller test. It is clear that by looking at the p values, we reject the null hypothesis, therefore we can conclude that the mean and variance of the rainfall pattern is constant over time.

Fitting the Logistic Regression Model

We aimed to create a model that predicts if rainfall will occur or not. First we created a new column for rain occurrence (0:No Rain, 1:Rain) which will be our dependent variable. Due to multicollinearity we have dropped following variables:

```
'apparent_temperature_max',
'apparent_temperature_min',
'et0_fao_evapotranspiration',
'temperature_2m_max', 'temperature_2m_min',
'temperature_2m_mean', 'windgusts_10m_max'
```

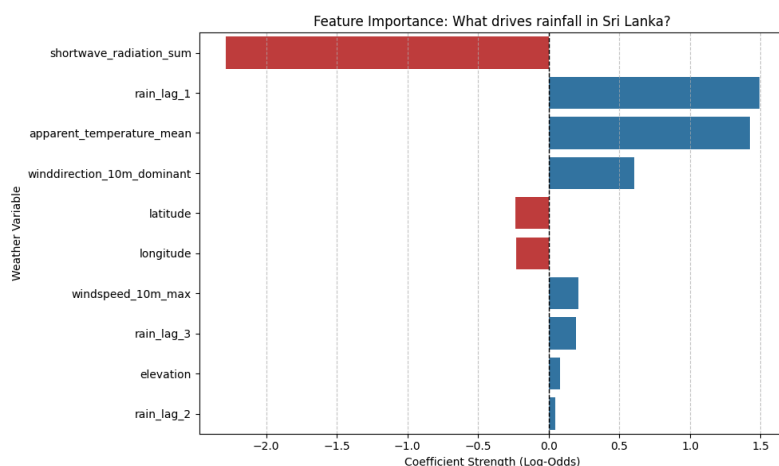
The dataset was split into 80% training and 20% testing sets, stratified by rain_occurrence to maintain class balance. RobustScaler was applied to numerical features to handle outliers and scale the data appropriately for logistic regression and a Logistic Regression model with `class_weight='balanced'` (to address class imbalance in rain occurrence) was trained.



We also created 3 columns for the 1st, 2nd and 3rd lag of the data (lags were taken making sure they came from the same city) to check the impact of how the previous day's weather can predict today's weather.

Model Performance Metrics:				
	precision	recall	f1-score	support
0	0.50	0.85	0.63	5338
1	0.96	0.81	0.88	24140
accuracy			0.82	29478
macro avg	0.73	0.83	0.75	29478
weighted avg	0.88	0.82	0.83	29478

The model achieves an overall accuracy of 82%. There's a relatively high number of False Negatives for rain-50% (i.e., the model predicted no rain, but it did rain), but the number of false positives is minimal. Overall the model is extremely good at predicting rain occurrence, however its prediction of rain not occurring is not as accurate.



By checking the feature importance it is clear that the shortwave radiation, rain lag and mean apparent temperature play a significant role. Therefore, as sunlight increases, the probability of rain drops drastically.

Lag 1 has a very strong positive coefficient. If it rained yesterday, the atmospheric conditions (humidity, pressure) often remain similar, making it highly likely to rain again today.

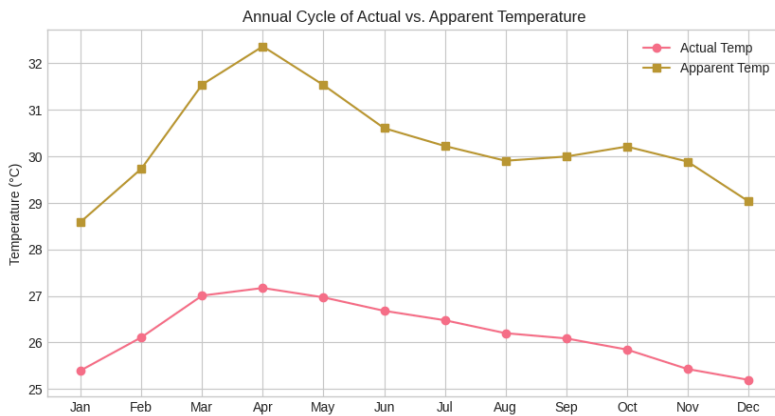
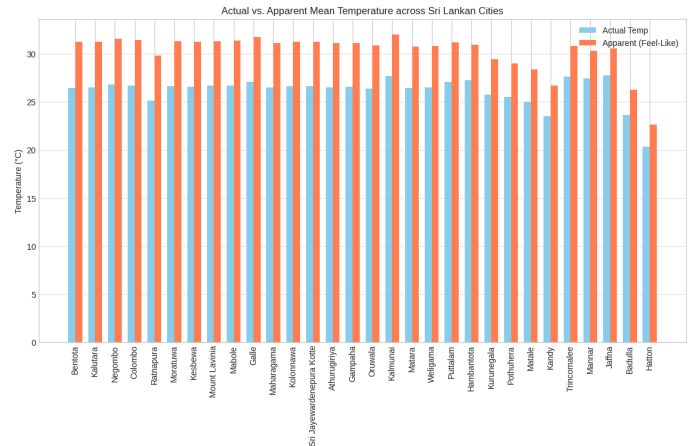
Apparent temperature mean- strongly increases the odds of rain. Apart from these, wind direction also plays a significant role.

4.2. Objective 2: Modeling Apparent Temperature and Thermal Comfort

Apparent temperature measures how heat is perceived by humans. Apparent Temperature. It incorporates humidity, wind speed, and atmospheric conditions and provides a more realistic measure of thermal comfort. This study aims to identify the environmental factors that influence perceived heat in Sri Lanka,

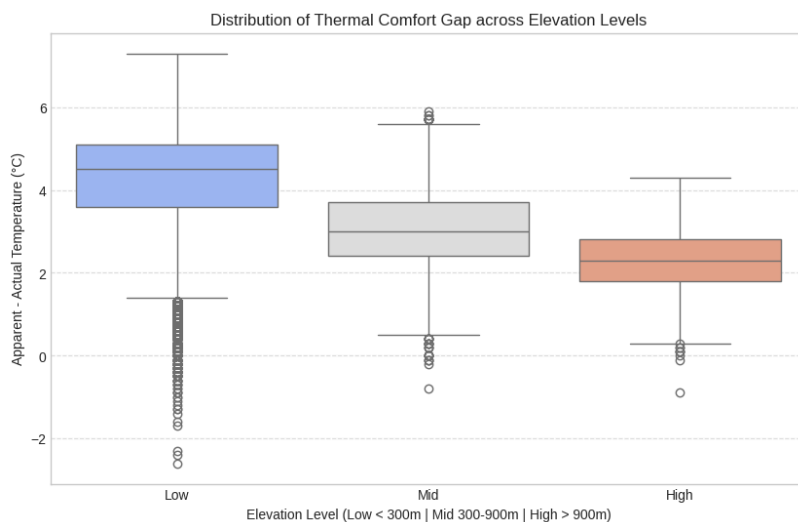
This bar chart is a comparison between temperature_2m_mean and apparent_temperature_mean of each city in the dataset and it reveals a consistent heat index effect across the country.

Coastal cities such as Colombo and Trincomalee exhibit Apparent Temperatures that are typically 2–4 °C higher than air temperatures due to high humidity. High-altitude locations such as Hatton display a narrower temperature gap.



Time series analysis reveals clear seasonal patterns in thermal perception. During inter-monsoon periods (March–April and October–November), low wind speeds and high humidity intensify heat perception, widening the temperature gap. During the Southwest Monsoon, stronger winds promote evaporative cooling, reducing perceived heat despite high ambient temperatures.

This boxplot shows that the thermal gap decreases as elevation increases.



Lowland areas have the largest median gap due to high humidity, while highland regions have a smaller gap because cooler temperatures and stronger winds reduce heat discomfort. The Lowlands also display greater variability and more extreme values, indicating occasional intense heat events, whereas the Highlands show more stable conditions.

apparent_vs_actual.ipynb

```
--- ANOVA ---
Null Hypothesis (H0):
The mean difference between Apparent and
Actual Temperature is equal across all elevation groups.
Alternative Hypothesis (H1):
At least one elevation group has a significantly
different mean 'thermal gap' compared to the others.

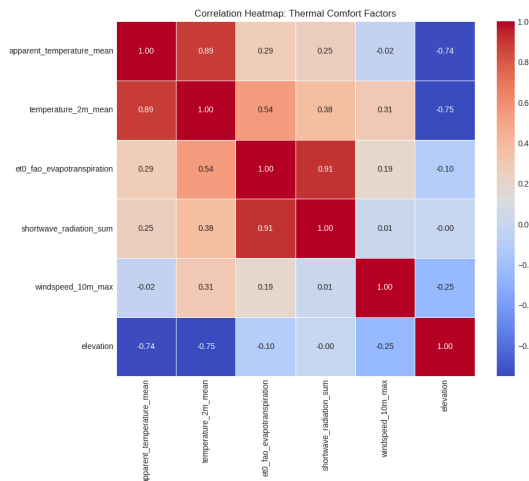
F-Statistic: 11263.69
P-Value: 0.0000e+00
Result: Reject the Null Hypothesis.
```

Post-hoc analysis confirmed that elevation significantly affects perceived temperature. Higher elevations experience lower moisture retention and air density, resulting in reduced heat stress compared to lowland and coastal regions

To examine whether elevation influences thermal comfort, an ANOVA test was also conducted. The test produced a statistically significant p-value ($p < 0.05$). Therefore, the null hypothesis was rejected.

Tukey HSD

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
High Low 1.9863 0.0 1.9431 2.0295 True
High Mid 0.7621 0.0 0.7131 0.8111 True
Low Mid -1.2242 0.0 -1.25 -1.1983 True
```



The correlation heatmap shows that Apparent Temperature is most strongly related to air temperature ($r = 0.89$), making it the main predictor. Elevation has a strong negative correlation ($r = -0.74$), indicating that higher-altitude areas generally feel cooler. Evapotranspiration (ET0) and shortwave radiation have moderate positive relationships with temperature, suggesting that humidity and solar energy also contribute to perceived heat.

To assess multicollinearity, a Variance Inflation Factor (VIF) analysis was conducted.

ET0 showed a high VIF (10.29), reflecting its strong dependence on temperature and solar radiation. Shortwave radiation also showed moderate multicollinearity ($VIF = 7.80$), due to its close relationship with temperature. In contrast, wind speed and elevation had low VIF values, indicating that they provide independent information.

VIF Scores

Feature	VIF
et0_fao_evapotranspiration	10.299464
shortwave_radiation_sum	7.802685
temperature_2m_mean	4.816536
elevation	3.283808
windspeed_10m_max	1.263302

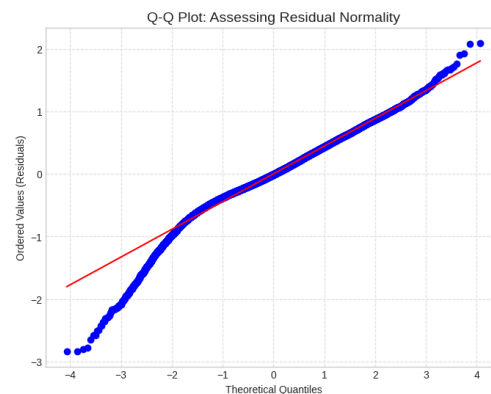
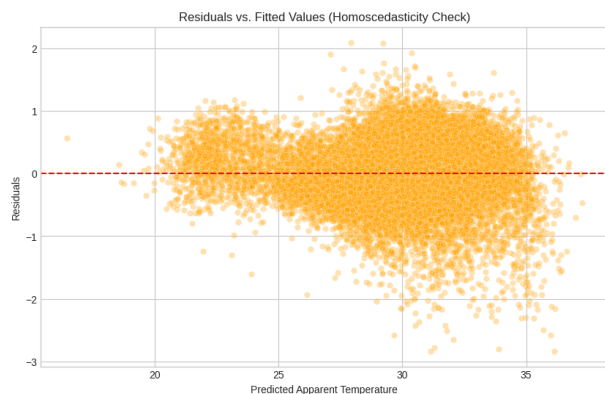
Model Performance

Model	R2	RMSE
With ET0	0.968600	0.447870
Without ET0	0.904654	0.780445

After VIF analysis two models were developed with ET0 and without ET0.

The results show that including ET0 improved performance, increasing R^2 from 0.9046 to 0.9686 and reducing RMSE from 0.7804 to 0.4478. Although ET0 had a high VIF value, the improvement in accuracy indicates that it provides important information about moisture related heat discomfort. Excluding ET0 would have reduced the model's ability to capture thermal conditions. Therefore, the enhanced model with ET0 was selected as the final model.

To assess the reliability of the model residual diagnostics were examined. Although the model showed good predictive performance the plots revealed violations of key linear regression assumptions.



Residuals vs. Fitted plot displayed a funnel-shaped pattern, indicating heteroscedasticity. This means that prediction errors increase at higher Apparent Temperatures, suggesting that the model becomes less reliable during extreme heat conditions and that the relationships between variables are partly non-linear.

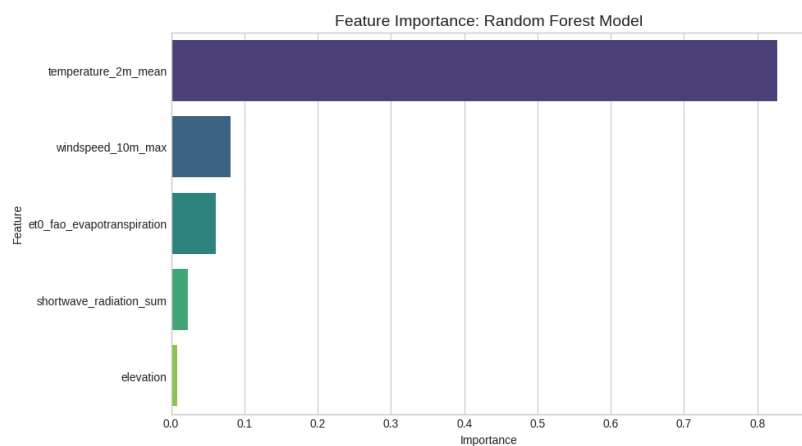
The Q–Q plot showed deviations from the diagonal line in the tails, indicating that residuals are not perfectly normally distributed. This suggests that the model struggles to capture extreme weather events, likely due to complex climatic effects that cannot be fully explained by a simple linear structure.

Although the initial Multiple Linear Regression model achieved a high R^2 score, residual and Q–Q plots showed violations of linearity and constant variance. This indicated that the relationships between weather variables were too complex for a simple linear model. To address this, a Random Forest Regressor was applied, which can naturally model non-linear interactions without manual transformations. This approach improved prediction accuracy, increased R^2 to near-perfect levels(0.9860), reduced RMSE(0.2989), and resolved the diagnostic issues observed in the linear model.

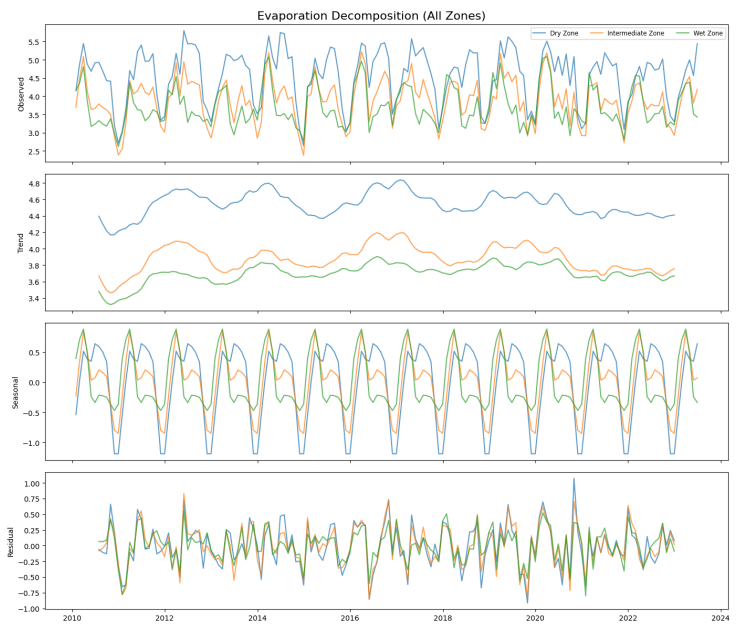
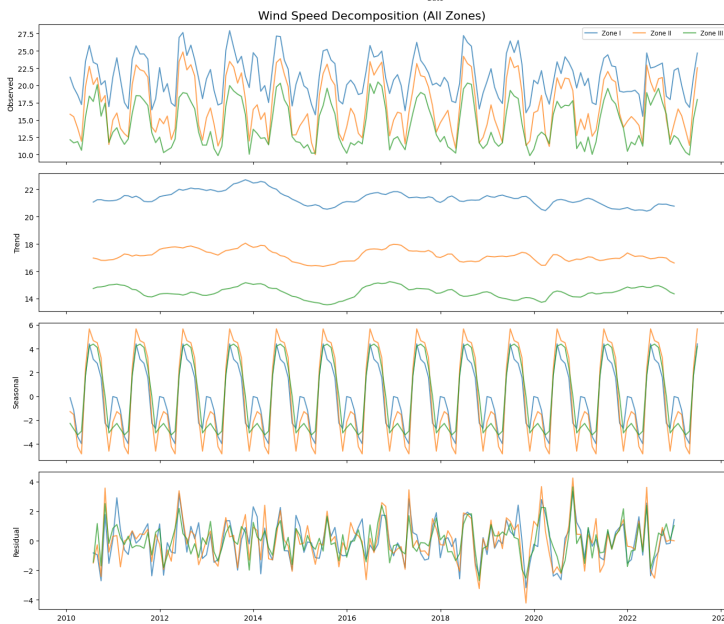
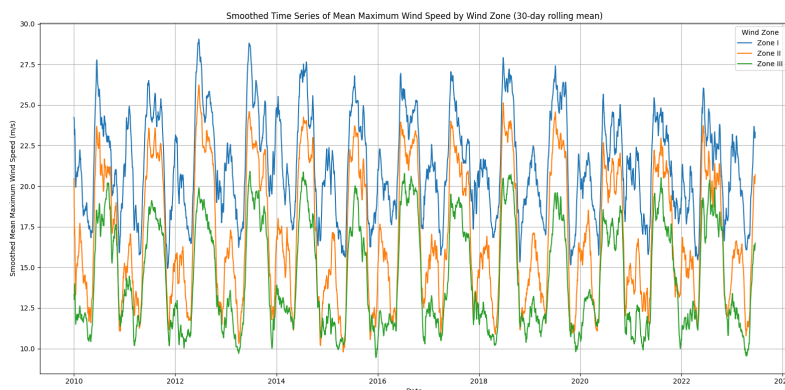
Random Forest Regression

Random Forest R^2 Score: 0.9860

Random Forest RMSE: 0.2989 °C



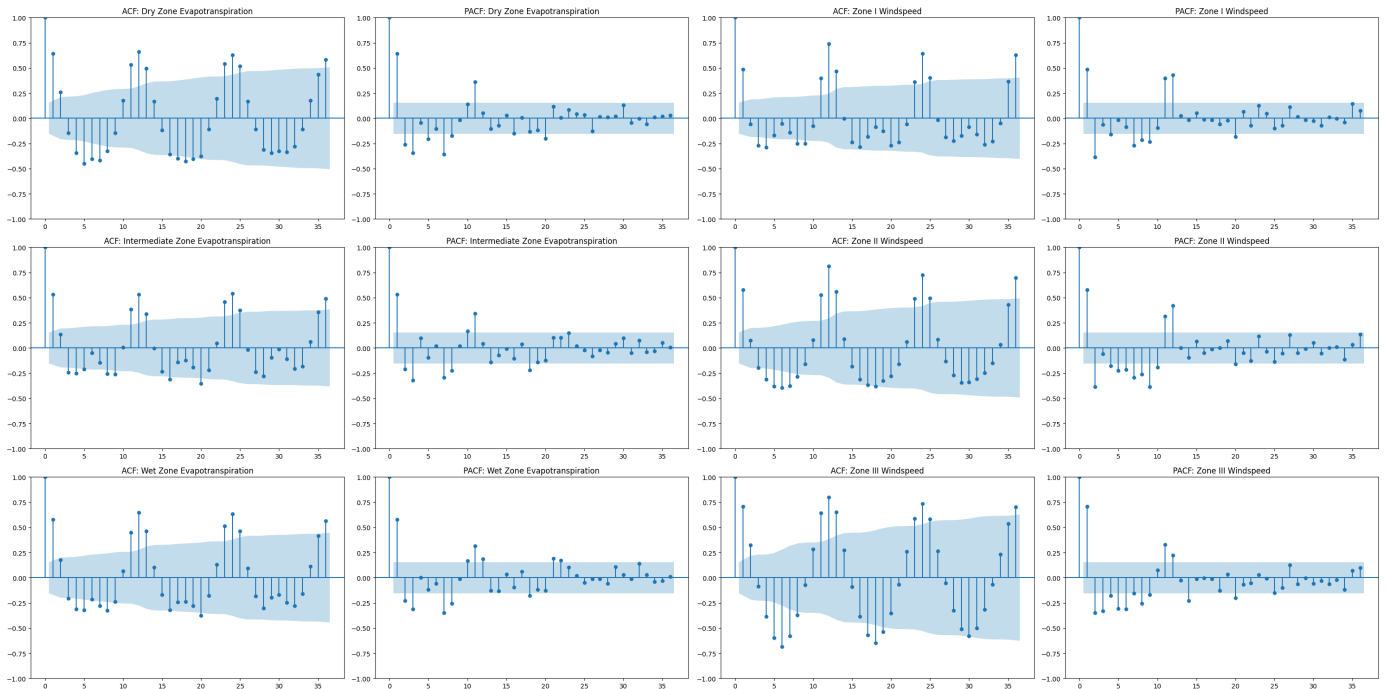
4.3. Objective 3: Analyzing Wind Dynamics and Evapotranspiration



Analysis of the time series data for daily evaporation and wind speeds reveals three distinct patterns for each variable. Evaporation patterns are categorized geographically, corresponding to the Dry Zone, Wet Zone, and Intermediate Zone. Wind speed data, however, is categorized based on the measured wind speeds. Future analyses of Evaporation and Wind speed patterns will be conducted separately for these zones. The Dry Zone consistently shows the highest ET₀ peaking near 5.5–5.8, and the Wet Zone the lowest around 4.5–5.0. This is expected, as higher temperatures and lower humidity in dry zones increase evapotranspiration. Zone I is the most volatile, with wind speeds frequently exceeding **25m/s**. Zone III is the most stable, generally staying below **20m/s**.

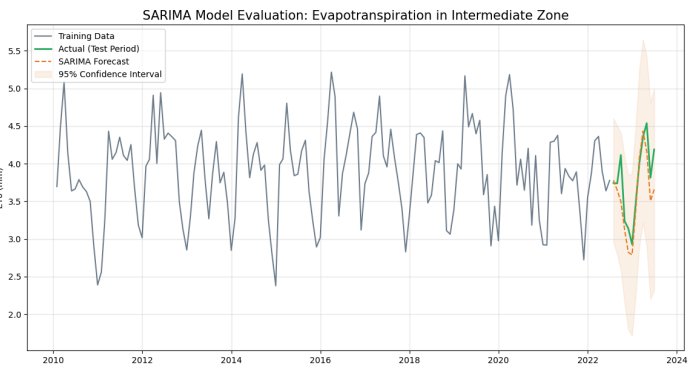
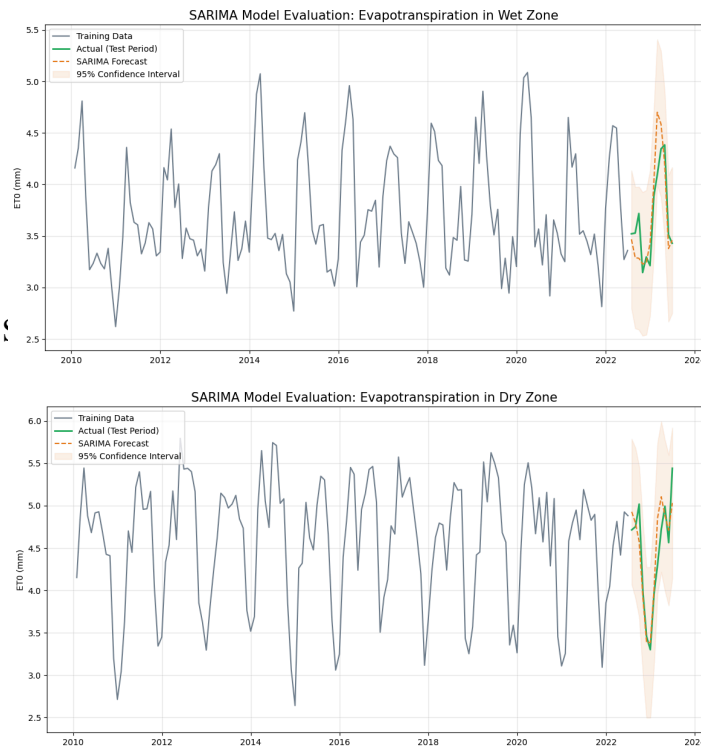
The seasonal decomposition across all three zones shows a consistent annual cycle. The peaks and troughs occur with high regularity, corresponding to dry and monsoon seasons. Multi-year oscillations are shown in the trend lines for all zones, meaning they are not linear. Specifically, a peak occurred around 2016–2017, and a noticeable dip around 2011. Residuals are mostly centered around zero. All three zones show a clear, stable annual seasonal wind speed pattern. Zone II has the largest seasonal amplitude, and Zone III is the most regular. Trends show multi-year variability with a common dip around 2015–2016 followed by stabilization. Zone I maintains the highest mean speeds and a stable trend. Zone II has the greatest variability and a significant residual noise, implying higher sensitivity. Zone III has the lowest mean speeds and the cleanest decomposition.

Variable	ADF_p	ADF_Stat	ADF_Result	KPSS_p	KPSS_Result
Evap - Dry Zone	0.0269	-3.0957	Stationary	0.1	Stationary
Evap - Intermediate Zone	0.0561	-2.8155	Non-Stationary	0.1	Stationary
Evap - Wet Zone	0.0643	-2.7595	Non-Stationary	0.1	Stationary
Wind - Zone I	0.2085	-2.1937	Non-Stationary	0.1	Stationary
Wind - Zone II	0.0591	-2.7945	Non-Stationary	0.1	Stationary
Wind - Zone III	0.0414	-2.9351	Stationary	0.1	Stationary



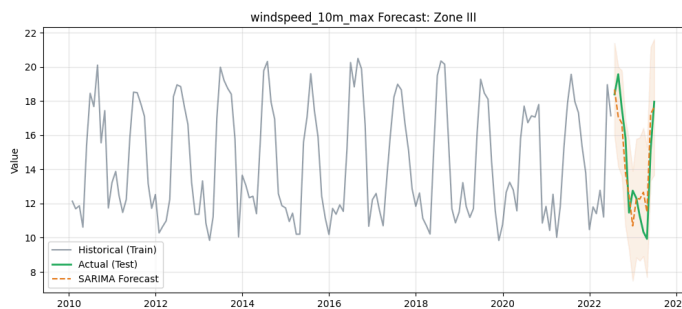
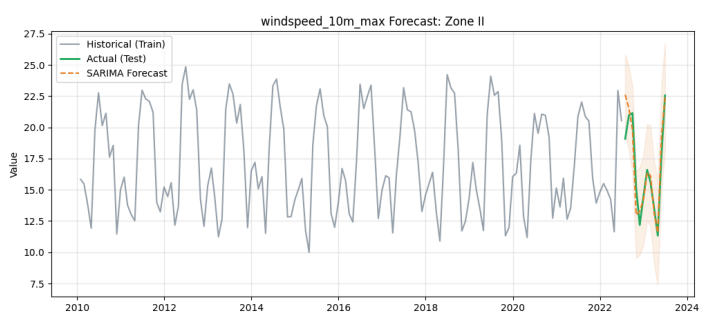
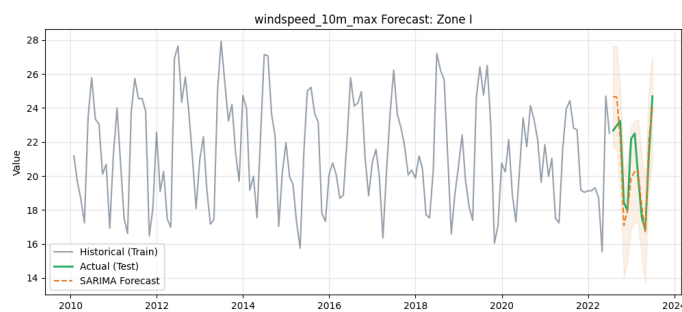
While the KPSS test consistently indicated stationarity, the Augmented Dickey-Fuller (ADF) test suggested non-stationary behavior, especially in the Wet and Intermediate zones. This discrepancy implies that the time series lack long-term stochastic drift but are heavily influenced by deterministic seasonal cycles. This strong seasonality was further confirmed by the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots. These plots exhibited significant sinusoidal spikes at 12, 24, and 36-month intervals, which is a definitive signature of Sri Lanka's dual monsoon system. Given the sharp cutoff observed at Lag 1 in the PACF, combined with the persistent seasonal peaks in the ACF, the SARIMA model can be used.

Zone	Best Model	AIC	MAE	RMSE	MAPE
Wet Zone	(1, 1, 3) x (2, 1, 0, 12)	116.311	0.2009 mm	0.2588 mm	5.31%
Intermediate Zone	(2, 1, 1) x (2, 1, 1, 12)	160.008	0.2315 mm	0.2970 mm	6.02%
Dry Zone	(1, 1, 1) x (2, 1, 0, 12)	181.401	0.2157 mm	0.2771 mm	4.68%



The Intermediate Zone required a more complex model, while the Dry Zone used a simpler model and showed the most stable behavior. All three models performed well, with MAPE values below 10%, and the Dry Zone giving the best MAPE of 4.68%. The small difference between RMSE and MAE shows that the prediction errors are consistent and not affected by large outliers. The forecast plots also show that the models closely follow the actual evapotranspiration values, correctly capturing seasonal patterns and providing reliable predictions.

Zone	Best Model	AIC	MAE	RMSE	MAPE
Zone I	(1, 0, 0) x (2, 0, 1, 12)	609.095	0.7965 m/s	0.9778 m/s	3.79%
Zone II	(1, 0, 0) x (1, 0, 1, 12)	611.674	0.7645 m/s	0.9811 m/s	4.62%
Zone III	(0, 0, 0) x (1, 0, 1, 12)	543.622	0.8737 m/s	0.9952 m/s	6.52%

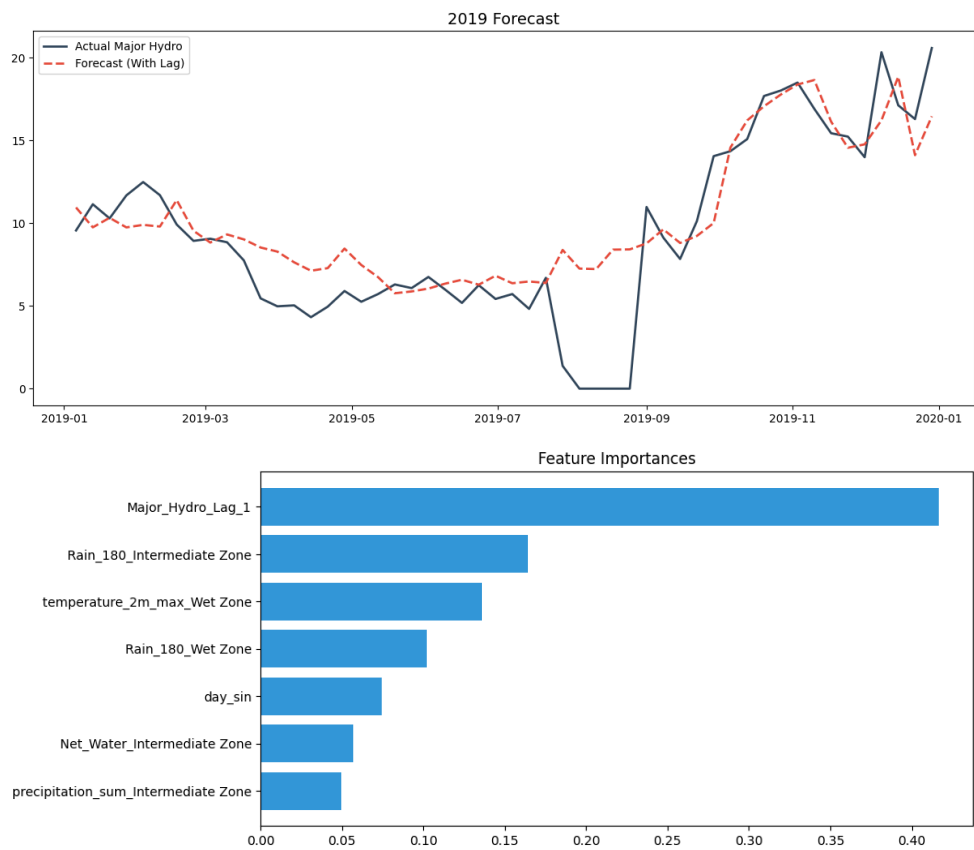


SARIMA models were developed to forecast monthly wind speed for Zone I, Zone II, and Zone III. Zone I and Zone II used similar models with both non-seasonal and seasonal components, while Zone III required a simpler model with only a seasonal component, indicating more regular seasonal behavior. The performance metrics show that all models provide good accuracy, with MAPE values of 3.79% for Zone I, 4.62% for Zone II, and 6.52% for Zone III. Zone I shows the best forecasting

performance with the lowest error values. The small differences between MAE and RMSE across all zones suggest that prediction errors are consistent and not affected by extreme values. The forecast plots further confirm that the SARIMA models closely follow the actual wind speed patterns, successfully capturing seasonal variations and providing reliable short-term forecasts.

A Random Forest model was utilized to forecast major hydroelectric power generation in Sri Lanka, capitalizing on the predictable nature of local evaporation patterns and other weather variables. For model development, data on major hydro power generation from 2016 to 2018 was used for training, while actual 2019 data was reserved for evaluation. The 2020-2023 major hydro data was excluded due to irregularities in the power generation figures, which were influenced by the financial crisis in Sri Lanka during that period.

	Test Set	Train Set
R-Squared	0.704	0.792
MAE	1.55 GWh/day	1.87 GWh/day
MAPE	27.85%	19.91%



The developed Random Forest model offers a robust and reliable methodology for forecasting major hydro power generation in Sri Lanka. It achieved R-squared of 0.704 when tested on 2019 data, a figure that is supported by the minimal difference between the training and testing scores, confirming the model's robustness and lack of overfitting. The forecast was typically off by only 1.55 GWh per day (MAE), which represents a reasonable margin of error given the volatile nature of weather and grid demand. While the

average percentage error (MAPE) of approximately 28% indicates that some daily fluctuations are likely driven by human operational decisions rather than weather. The model works well because it combines how full reservoirs are over time, with the amount of power made the week before. This helps it understand how rivers naturally flow and move. Feature analysis identified accumulated rainfall and hydro power generated in recent history as the strongest drivers of power output.

4.4. Objective 4: Geographical Influences on Extreme Events in Climate Variables

Extreme climate events were analyzed using statistical anomaly detection techniques to identify unusual patterns in key climate variables across different geographical locations. The primary variables considered were mean 2 m air temperature, precipitation sum, shortwave radiation sum, and maximum 10 m wind speed.

Z-score (and MAD Score) based anomaly detection identified numerous extreme events across key climate variables. The distribution of daily mean temperature is clearly not normal exhibiting skewness driven by strong geographical heterogeneity. As a result, the Z-score method substantially overestimates anomalies, particularly in regions with distinct local climates. For example, naturally cooler high-altitude locations such as Hatton are systematically misclassified as anomalous because their normal temperature range falls far below the global mean, leading to a large number of false positives. This univariate approach also ignores interactions among climate variables, treating each variable in isolation and failing to account for compound climatic behavior influenced by elevation, coastal exposure, and monsoonal processes. Consequently, while simple to implement, the Z-score method proves inadequate for accurately identifying meaningful extreme events in geographically diverse climate datasets.

anomaly_detection.ipynb

Anomalies in temperature_2m_mean (Z-Score > 3): 2864

	time	city	temperature_2m_mean
45910	2014-07-25	Kalmunai	31.8
137079	2021-11-26	Hatton	20.6
98179	2023-01-28	Badulla	20.6
94863	2013-12-30	Badulla	20.6
135761	2018-04-18	Hatton	20.6

Anomalies in shortwave_radiation_sum (Z-Score > 3): 1318

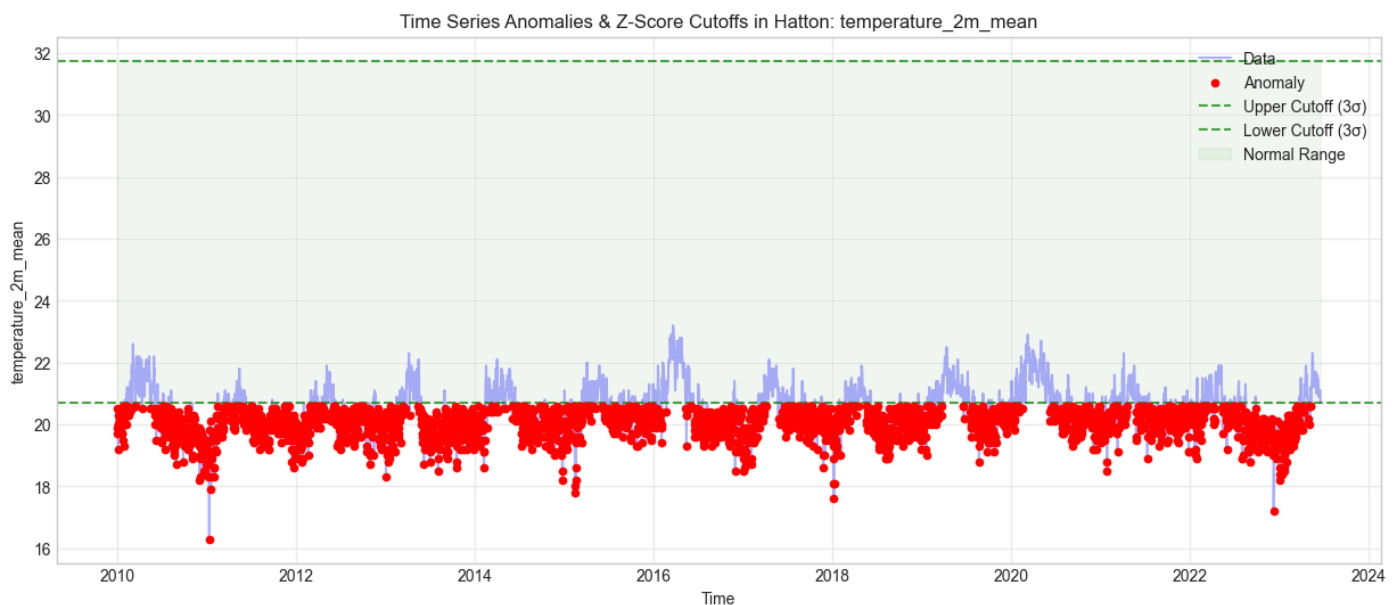
	time	city	shortwave_radiation_sum
114502	2013-12-05	Mannar	5.86
133781	2012-11-15	Hatton	5.86
51284	2015-10-26	Trincomalee	5.86
129634	2014-12-24	Mabole	5.85
85390	2014-12-24	Gampaha	5.85

Anomalies in precipitation_sum (Z-Score > 3): 2096

	time	city	precipitation_sum
120310	2016-05-15	Pothuhera	338.8
125226	2016-05-15	Kurunegala	338.8
110478	2016-05-15	Matale	300.4
144890	2016-05-15	Oruwala	261.8
85898	2016-05-15	Gampaha	261.8

Anomalies in windspeed_10m_max (Z-Score > 3): 1298

	time	city	windspeed_10m_max
62785	2020-05-21	Jaffna	50.2
63154	2021-05-25	Jaffna	48.5
63717	2022-12-09	Jaffna	48.3
63524	2022-05-30	Jaffna	48.0
59129	2010-05-18	Jaffna	48.0



anomaly_detection.ipynb

Total Anomalies Detected by Isolation Forest: 1180

Sample Anomalies:

	time	city	temperature_2m_mean	shortwave_radiation_sum	\
62784	2020-05-20	Jaffna	30.4	25.00	
61682	2017-05-14	Jaffna	30.3	23.70	
62398	2019-04-30	Jaffna	30.2	11.43	
61322	2016-05-19	Jaffna	29.4	20.74	
59132	2010-05-21	Jaffna	29.3	11.35	

	precipitation_sum	windspeed_10m_max	elevation	anomaly
62784	0.0	41.2	5	-1
61682	0.2	40.3	5	-1
62398	0.8	40.0	5	-1
61322	10.0	45.5	5	-1
59132	3.8	40.3	5	-1

To address the limitations of univariate statistical methods, a multivariate Isolation Forest algorithm was applied, identifying 1,475 anomalous events across the combined climate variables. The superiority of this approach is evident from the underlying data distributions, like the histogram of daily mean temperature is bimodal and left-skewed rather than normally distributed, with a primary peak around 26 °C and a secondary cluster near 20 °C. This violates the core assumption of the Z-score method, which relies on a single global mean (26.23 °C)

and consequently produces a large number of false positives, particularly in naturally cooler regions such as Hatton. Time-series analysis for Hatton demonstrates that temperatures around 20 °C represent normal climatic conditions rather than extremes, yet are incorrectly flagged by the Z-score method.

Several classification models were initially trained and evaluated in their standard (unbalanced) form to assess their ability to detect climate anomalies, including Logistic Regression, Random Forest, Gradient Boost and XGBoost.

Confusion-matrix analysis shows that these models achieve high precision but relatively low sensitivity to extreme events, for example, the standard XGBoost correctly identified 212

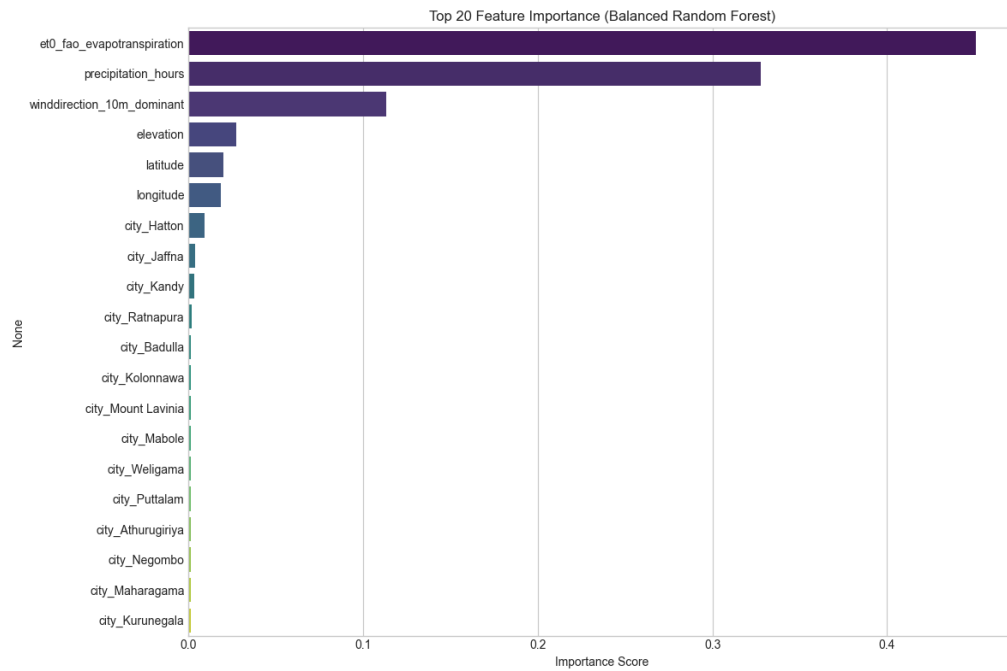
anomalies while missing only 87, while the others missed many of true anomalies, this indicates the existence of severe class imbalance due to the fact that XGboost is known to be able to handle class imbalances . Although false alarms were minimal in this stage, the high number of false negatives indicated that many extreme events were not being detected.

	TP (Caught)	FN (Missed)	FP (False Alarms)	TN (Correct 0s)
Model				
LogReg	108	191	14	29,183
RandomForest	151	148	45	29,152
GradientBoosting	155	144	41	29,156
XGBoost	183	116	59	29,138

	TP (Caught)	FN (Missed)	FP (False Alarms)	TN (Correct 0s)
Model				
LogReg	108	191	14	29,183
RandomForest	151	148	45	29,152
GradientBoosting	155	144	41	29,156
XGBoost	183	116	59	29,138
LogReg (Balanced)	268	31	3,166	26,031
Elastic Net (Balanced)	266	33	4,406	24,791
SMOTE + LogReg	265	34	3,055	26,142
SMOTEENN + LogReg	267	32	3,270	25,927
RandomUnderSampler + LogReg	270	29	3,323	25,874
Balanced Random Forest	272	27	1,438	27,759
Easy Ensemble	267	32	1,885	27,312

Given the strong class imbalance in the dataset and the critical importance of capturing rare extreme events, a second modelling stage was undertaken using imbalance-aware techniques. Balanced ensemble models and resampling-based linear methods substantially improved anomaly detection compared to the initial unbalanced models. In particular, the Balanced Random Forest detected 287 of

the 299 anomalies, missing only 12 events, while the Easy Ensemble identified 281 anomalies with similarly low miss rates. Sampling and class-weighted Logistic Regression variants also achieved high recall, typically capturing over 270 anomalies, though at the cost of a larger number of false positives. These results clearly illustrate the fundamental trade-off introduced by imbalance handling: a sharp reduction in missed extreme events is accompanied by an increase in false alarms.



The feature importance plot reveals that the Balanced Random Forest model is overwhelmingly driven by three variables: evapotranspiration, precipitation hours, and wind direction which collectively dwarf the influence of geographical coordinates and specific city locations.

By refactoring the model to focus exclusively on these 3 features, we see that the Balanced Random Forest remains a top performer. Focusing on just the top three features evapotranspiration, precipitation hours, and wind direction has led to a slight increase in False Alarms (FP) although the top three variables carry the most weight, the minor features removed assisted in prediction.

	TP (Caught)	FN (Missed)	FP (False Alarms)	TN (Correct 0s)
Model				
LogReg	102	197	14	29,183
RandomForest	185	114	34	29,163
GradientBoosting	147	152	37	29,160
XGBoost	181	118	49	29,148
LogReg (Balanced)	261	38	3,759	25,438
Elastic Net (Balanced)	256	43	6,321	22,876
SMOTE + LogReg	261	38	3,775	25,422
SMOTEENN + LogReg	261	38	3,845	25,352
RandomUnderSampler + LogReg	261	38	3,862	25,335
Balanced Random Forest	263	36	1,521	27,676
Easy Ensemble	248	51	1,640	27,557

Ultimately, the streamlined model trades a marginal amount of statistical precision for a significant gain in simplicity and interpretability, maintaining a high True Positive count while relying on a much more transparent set of physical drivers.

5. Conclusion

This comprehensive analysis of Sri Lanka's weather patterns from 2010 to 2023 highlights the critical role of statistical modeling and machine learning can play in understanding the island's climate. By integrating meteorological data with predictive modeling, this study offers significant insights for water resource management, energy planning, and thermal comfort assessment.

Key findings from the study include:

- **Precipitation Dynamics:** Rainfall in Sri Lanka is highly seasonal, with the second inter-monsoon period (October–November) being the wettest. The application of Logistic Regression proved highly effective (82% accuracy) in predicting rainfall occurrence, identifying shortwave radiation and previous-day rainfall as the strongest predictors.
- **Thermal Comfort & Non-Linearity:** The study confirmed that "Apparent Temperature" is a critical metric for Sri Lanka, driven largely by humidity in coastal regions. While linear models failed to capture the complex interactions of heat stress, the Random Forest Regressor successfully modeled these non-linear relationships ($R^2 \approx 0.98$), validating that elevation and evapotranspiration are key determinants of human comfort.
- **Resource Forecasting:** The strong seasonality observed in wind and evaporation rates allowed for accurate time-series forecasting using SARIMA models. Furthermore, the study demonstrated the viability of using Random Forest models to forecast major hydroelectric power generation ($R^2 = 0.704$), linking meteorological inputs directly to energy grid potential.
- **Extreme Event Detection:** The analysis revealed that traditional statistical methods (like Z-scores) are inadequate for Sri Lanka's non-normal, geographically diverse climate data. Multivariate Isolation Forests and Balanced Random Forests proved superior in detecting extreme weather anomalies, highlighting the necessity of handling class imbalances when modeling rare climate events.

Ultimately, this study demonstrates that while Sri Lanka's climate is influenced by predictable monsoonal cycles, the intensity of specific variables requires robust, non-linear machine learning approaches. These models provide a scalable foundation for predicting environmental shifts, optimizing renewable energy production, and mitigating climate-related risks in the future.